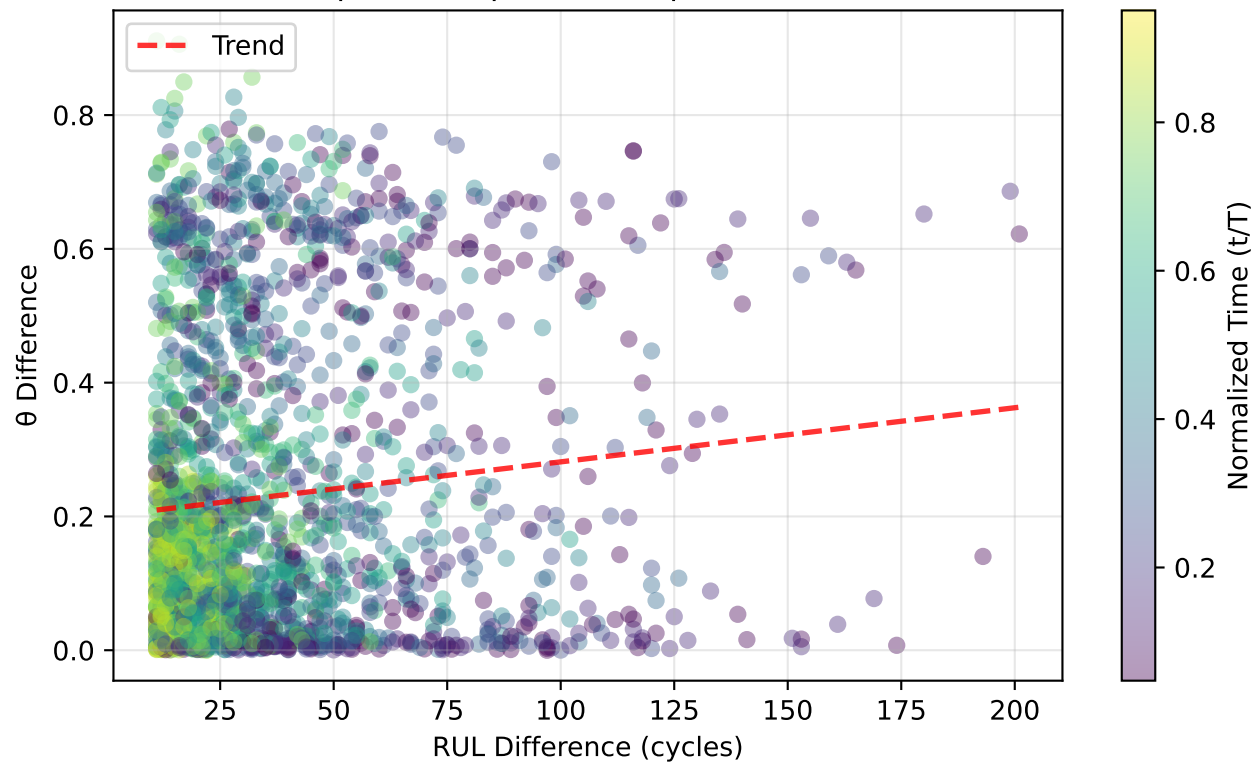
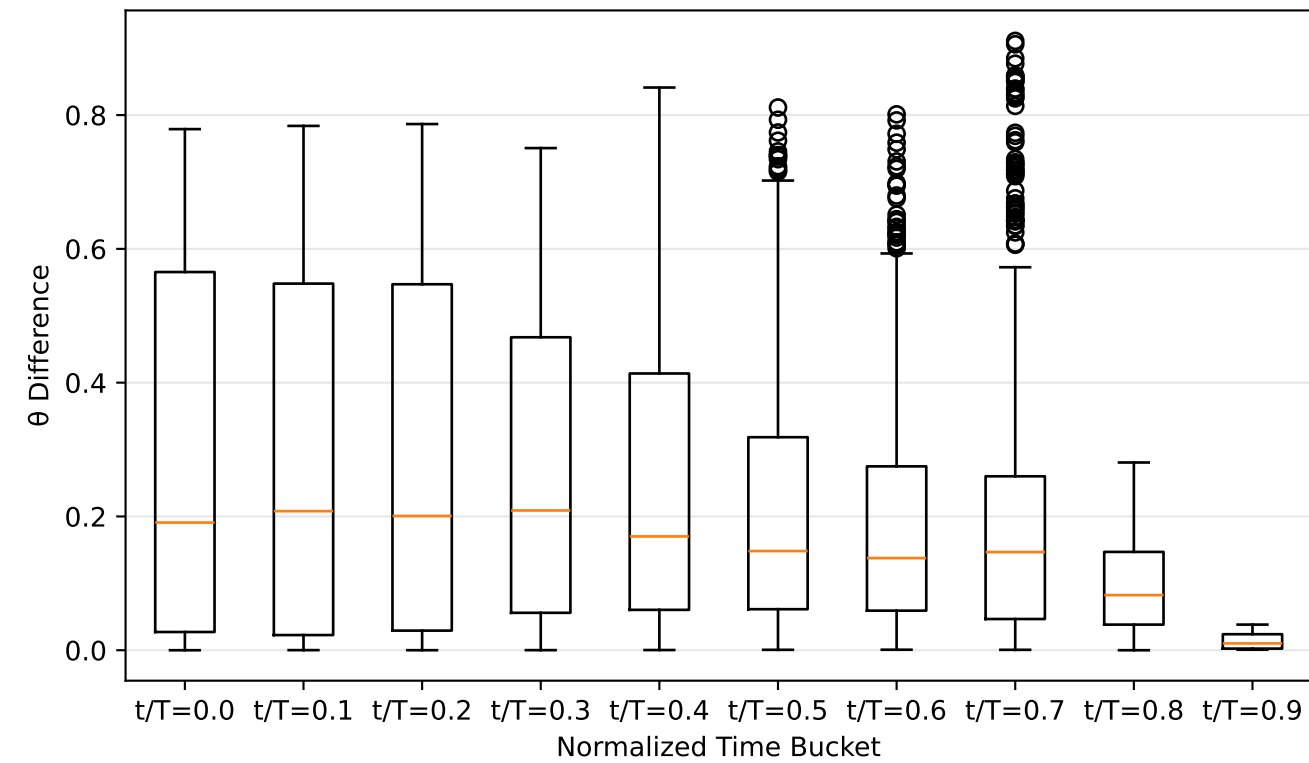


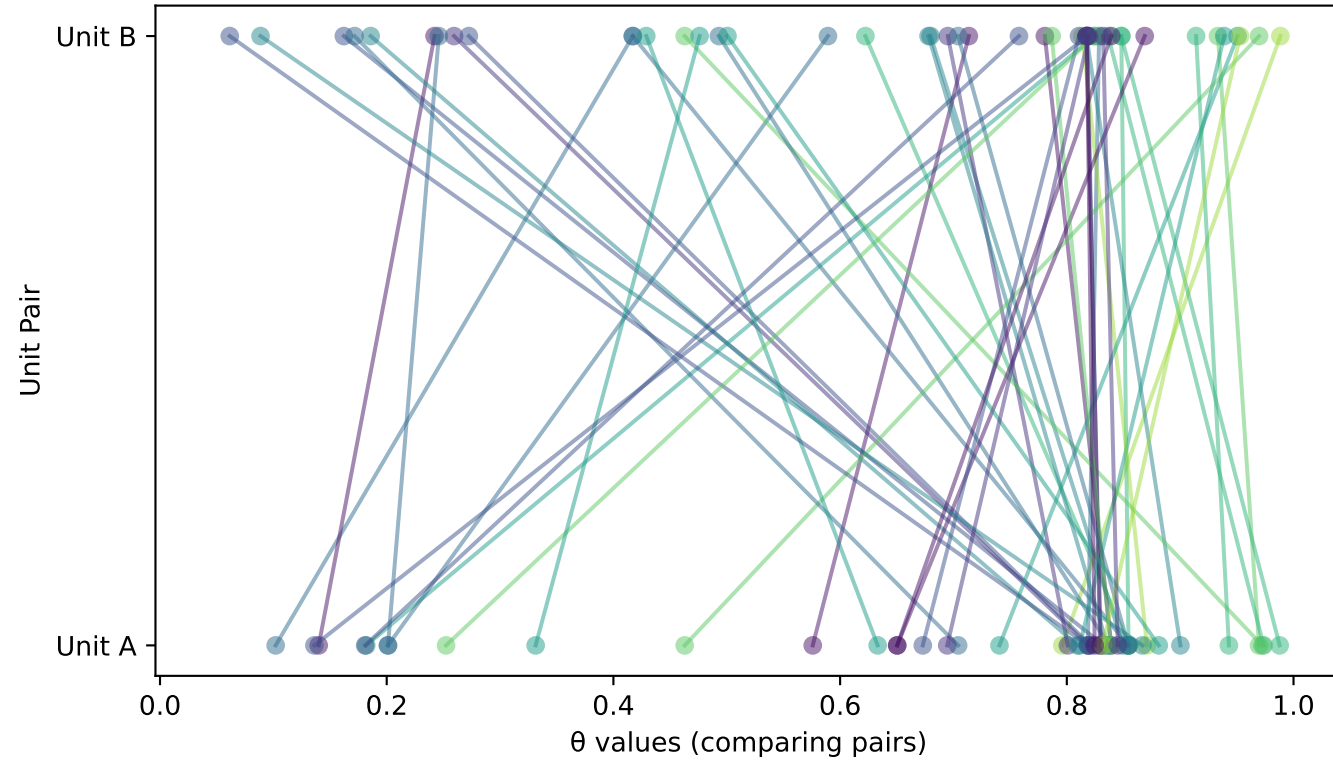
Health State Difference vs θ Difference
(Spearman $\rho=0.0827$, $p=2.54e-08$)



θ Difference Across Different Times



Sample Pairs: Showing θ Differences Within Pairs



Test 1c: Same-Time Different-Health Analysis

Dataset: FD001
Latent Dim: 4
Window Size: 10

Results:

Comparison Pairs: 4520

RUL Difference:
Mean: 36.0 cycles
Median: 27.0 cycles
Max: 201.0 cycles

θ Difference:
Mean: 0.2298
Median: 0.1449
Max: 0.9109

Correlation:
Spearman $\rho = 0.0827$
 p -value = $2.54e-08$

Interpretation:

If $\rho \gg 0$ and $p < 0.05$:
→ θ TRACKS STATE (different health leads to different θ even at same t/T)

If $\rho \approx 0$ or $p > 0.05$:
→ θ may be learning just t/T (health difference doesn't affect θ)